M.SHAHZAIB

FULL STACK DEVELOPER

RWP | mrshahzaib2505@gmail.com | www.linkedin.com/in/mr-shahzaib | https://github.com/MrShahzaib005 | 03320049989

SUMMARY

Software engineer and BSCS student with hands-on experience in robotics software (ROS 2) and full-stack web development. Built the Arbotrix company website and developed ROS 2 robot platforms, integrating embedded systems with web interfaces and APIs. Strong problem-solving skills, backend and database experience, and a collaborative approach to delivering scalable solutions. Seeking internships or junior roles in robotics software, backend, or full-stack development.

PROFESSIONAL EXPERIENCE

Web Developer (Lead) | Arbotrix · Islamabad, Pakistan | **arbotrix.com**

- Lead developer for the company website (arbotrix.com) built with Next.js and React implemented product/service pages, course pages, sign-in/registration flows, and technical case studies.
- Developed responsive, SEO-friendly Next.js pages with server-side rendering and dynamic routing to improve discoverability and UX.
- Built secure authentication and user registration flows and integrated backend APIs for course access and account management.
- Optimized front-end performance and accessibility for faster load times and better mobile experience.
- Collaborated with designers and engineers to translate requirements into maintainable, scalable features and handled deployments.

KEY PROJECTS

Buddy AI — Agentic Assistant with Human-in-the-Loop Payments

- Built an LLM-agent backend that drafts transactional actions (food, e-commerce, flights) but requires explicit user approval before execution.
- Designed and enforced per-user configurable daily spending caps at approval time and implemented an approval flow linking chat to a transaction-summary page.
- Developed a Flutter client with a global AppState and an offline mock mode (USE_MOCK) so the UI team can work independently; roadmap includes JazzCash/Easypaisa payment gateway integration.

Arbotrix Web Application · Next.js, Tailwind CSS, Supabase · arbotrix.com

- Engineered a server-side rendered application utilizing Next.js App Router, Prisma ORM, and PostgreSQL for robust relational data management and secure server actions.
- Developed dynamic, conversion-optimized UI/UX funnels and integrated Supabase Auth for secure user sessions across technical case studies and course enrollment pages.

University E-Portal System | Java Spring Boot, MySQL

- Architected a comprehensive student management system featuring Role-Based Access Control (RBAC) for admins, teachers, and students.
- Developed RESTful APIs using Spring Boot to handle course enrollment, grade viewing, and attendance tracking with 99% uptime.
- Implemented secure authentication using Spring Security and JWT (JSON Web Tokens) to protect sensitive student data.

AI-Powered Chat Interface (Gemini Clone) | React.js, Google Gemini API

- Built a responsive conversational AI interface integrating the Google Gemini API for real-time natural language processing.

- Optimized state management using React Hooks (useContext/Reducers) to handle chat history and maintain context across sessions.
- Designed a modern, dark-mode UI with CSS Modules, ensuring a seamless user experience similar to production-level SaaS tools.

Dodo Bot: mobile app and ROS 2 robot

- Developed a mobile app to create Nav2 maps, send navigation goals, and visualize robot state for a differential-drive logistics robot.
- Implemented ROS 2 navigation (Nav2), EKF odometry, and SLAM on a Jetson Orin Nano with a custom URDF and LiDAR to enable reliable autonomous navigation.
- Enabled headless operation and remote maintenance over WiFi/SSH and integrated robot–app communication via ROS topics and services.

TECHNICAL SKILLS

Java (Spring Boot), JavaScript (ES6+), SQL, HTML5, CSS3

Frameworks & Libraries: React.js, Tailwind CSS, Bootstrap, Hibernate.

Tools: Git/GitHub, Postman, VS Code, IntelliJ, MySQL.

EDUCATION

Bachelor of Science in Computer Science **Feb 2023 - Sep 2027**

Iqra University H-9

- Current Semester: 6th
- Relevant Coursework: Web Engineering, Database Systems, Object-Oriented Programming, Data Structures & Algorithms.

CERTIFICATIONS & ACHIEVEMENTS

QS Impact Challenge | Certificate of Achievement

- Participated in a global challenge focused on solving muliple SDG's problems through innovation.

Game Development Workshop | Game Developers Conference (GDC)

- Completed intensive training on game logic and design patterns, applying concepts to interactive software development.